

Marco Storace

professor

基本信息

姓名: Marco Storace

所属院校: 热那亚大学

地区: Italy

年龄范围: 50岁以上

学科领域

自动化

电气工程及其自动化

神经科学

电子与计算机工程

系统科学与工程

集成电路设计与集成系统

研究领域

计算神经科学

电力电子与电机驱动

电力系统仿真与优化

自动控制理论与应用

脑科学与类脑智能

嵌入式系统与硬件设计

FPGA与可重构计算

系统辨识与优化

计算机控制系统与嵌入式

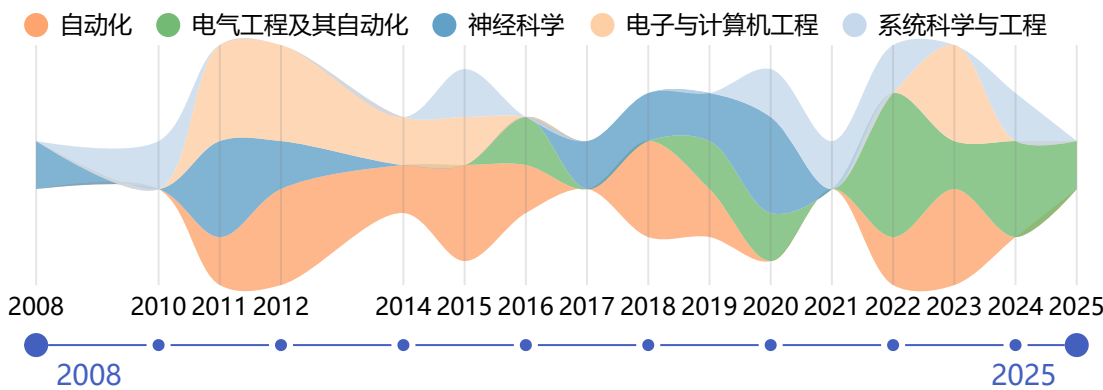
生物信号处理与模式识别

+6

个人简介

Marco Storace (M'01, SM '14) is a Full Professor at the University of Genoa, Italy, in the Department of Electrical, Electronic, Telecommunications Engineering and Naval Architecture. Born in 1969 in Genoa, he obtained his M.Sc. in Electronic Engineering in 1994 and Ph.D. in Electrical Engineering in 1998 from the same university. He visited EPFL in 1998 and 2002. His research focuses on nonlinear circuit theory and applications. He has authored or co-authored about 140 scientific papers and co-written a book. He served as an Associate Editor of IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS -- II and holds various positions in the IEEE Nonlinear Circuits and Systems Technical Committee. He speaks English, French, and Italian.

近年发表的论文



Modeling Amorphous-Core Inductors up to Magnetic Saturation

IEEE TRANSACTIONS ON POWER ELECTRONICS

Ravera, Alessandro; Oliveri, Alberto; Lodi, Matteo; Beatrice, Cinzia; Ferrara, Enzo; Fiorillo, Fausto; Storace, Marco

被引用次数: 2 2025

Modeling the Effect of Air-Gap Length and Number of Turns on Ferrite-Core Inductors Working up to Magnetic Saturation in a Buck Converter

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ravera, Alessandro; Formentini, Andrea; Lodi, Matteo; Oliveri, Alberto; Passalacqua, Massimiliano; Storace, Marco

被引用次数: 0 2024

Patterns of synchronized clusters in adaptive networks

COMMUNICATIONS PHYSICS

Lodi, Matteo; Panahi, Shirin; Sorrentino, Francesco; Torcini, Alessandro; Storace, Marco

被引用次数: 0 2024

A nonlinear model of air-gapped ferrite -core inductors for SMPS applications

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Ravera, Alessandro; Formentini, Andrea; Lodi, Matteo; Oliveri, Alberto; Storace, Marco

被引用次数: 0 2024

Bridging functional and anatomical neural connectivity through cluster synchronization

SCIENTIFIC REPORTS

Baruzzi, Valentina; Lodi, Matteo; Sorrentino, Francesco; Storace, Marco

被引用次数: 1 2023

Co-Design of a Controller and Its Digital Implementation: The MOBY-DIC2 Toolbox for Embedded Model Predictive Control

IEEE TRANSACTIONS ON CONTROL SYSTEMS TECHNOLOGY

Ravera, Alessandro; Oliveri, Alberto; Lodi, Matteo; Bemporad, Alberto; Heemels, W. P. M. H.; Kerrigan, Eric C.; Storace, Marco

被引用次数: 0	2023
Stability Analysis of the Synchronous Solution in Arrays of Memristive Chua's Circuits	
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS	
Lodi, Matteo; Forti, Mauro; Storace, Marco	
被引用次数: 1	2023
Use of a Partially Saturating Inductor in a Boost Converter with Model Predictive Control	
ELECTRONICS	
Firpo, Pietro; Ravera, Alessandro; Oliveri, Alberto; Lodi, Matteo; Storace, Marco	
被引用次数: 3	2023
Pinning control of networks: Dimensionality reduction through simultaneous block-diagonalization of matrices	
CHAOS	
Panahi, Shirin; Lodi, Matteo; Storace, Marco; Sorrentino, Francesco	
被引用次数: 3	2022
Nonlinear models of power inductors: A survey	
INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS	
Oliveri, Alberto; Lodi, Matteo; Storace, Marco	
被引用次数: 20	2022
Estimation of inertia in power grids with turbine governors	
2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)	
Baruzzi, Valentina; Lodi, Matteo; Oliveri, Alberto; Storace, Marco	
被引用次数: 0	2022
Behavioral model of an amorphous-core inductor working up to partial saturation	
2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)	
Oliveri, Alberto; Lodi, Matteo; Beatrice, Cinzia; Ferrara, Enzo; Storace, Marco; Fiorillo, Fausto	
被引用次数: 3	2022
Forget partitions? Not yet...	
2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)	
Lodi, Matteo; Sorrentino, Francesco; Storace, Marco	
被引用次数: 1	2022
One-way dependent clusters and stability of cluster synchronization in directed networks	
NATURE COMMUNICATIONS	
Lodi, Matteo; Sorrentino, Francesco; Storace, Marco	
被引用次数: 22	2021
Delays induced cluster synchronization in chaotic networks	
CHAOS	
Nathe, Chad; Huang, Ke; Lodi, Matteo; Storace, Marco; Sorrentino, Francesco	

被引用次数: 1	2020
Analyzing synchronized clusters in neuron networks	
SCIENTIFIC REPORTS	
Lodi, Matteo; Della Rossa, Fabio; Sorrentino, Francesco; Storace, Marco	
被引用次数: 21	2020
Design Principles for Central Pattern Generators With Preset Rhythms	
IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS	
Lodi, Matteo; Shilnikov, Andrey L.; Storace, Marco	
被引用次数: 24	2020
A Nonlinear Behavioral Ferrite-Core Inductance Model Able to Reproduce Thermal Transients in Switch-Mode Power Supplies	
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS	
Lodi, Matteo; Bizzarri, Federico; Linaro, Daniele; Oliveri, Alberto; Brambilla, Angelo; Storace, Marco	
被引用次数: 8	2020
Mode Based Compensation of Rate-Dependent Hysteresis in a Piezoresistive Strain Sensor	
IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS	
Oliveri, Alberto; Maselli, Martina; Lodi, Matteo; Storace, Marco; Cianchetti, Matteo	
被引用次数: 39	2019
A Power-Loss-Dependent Inductance Model for Ferrite-Core Power Inductors in Switch-Mode Power Supplies	
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS	
Oliveri, Alberto; Di Capua, Giulia; Stoyka, Kateryna; Lodi, Matteo; Storace, Marco; Femia, Nicola	
被引用次数: 32	2019
An MPC-Based Approach for Emergency Control Ensuring Transient Stability in Power Grids With Steam Plants	
IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS	
Bonfiglio, Andrea; Cantoni, Federico; Oliveri, Alberto; Procopio, Renato; Rosini, Alessandro; Invernizzi, Marco; Storace, Marco	
被引用次数: 23	2019
Dimensional reduction in networks of non-Markovian spiking neurons: Equivalence of synaptic filtering and heterogeneous propagation delays	
PLOS COMPUTATIONAL BIOLOGY	
Mattia, Maurizio; Biggio, Matteo; Galluzzi, Andrea; Storace, Marco	
被引用次数: 12	2019
Design of Synthetic Central Pattern Generators Producing Desired Quadruped Gaits	
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS	
Lodi, Matteo; Shilnikov, Andrey; Storace, Marco	
被引用次数: 14	2018

Application of a low-cost piezoelectric displacement estimation technique based on laser interferometry for hysteresis open-loop compensation in an AFM scanner

PHYSICA B-CONDENSED MATTER

Oliveri, Alberto; Lodi, Matteo; Storace, Marco; Raiteri, Roberto

被引用次数: 3

2018

Model Reduction for Optimized Online Compensation of Hysteresis and Creep in Piezoelectric Actuators

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Oliveri, Alberto; Lodi, Matteo; Parodi, Mauro; Stellino, Flavio; Storace, Marco

被引用次数: 11

2018

A Switched Predictive Controller for an Electrical Powertrain System With Backlash

IEEE TRANSACTIONS ON POWER ELECTRONICS

Formentini, Andrea; Oliveri, Alberto; Marchesoni, Mario; Storace, Marco

被引用次数: 36

2017

Phase analysis method for burst onset prediction

PHYSICAL REVIEW E

Stellino, Flavio; Mazzoni, Alberto; Storace, Marco

被引用次数: 0

2017

Two FPGA-Oriented High-Speed Irradiance Virtual Sensors for Photovoltaic Plants

IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS

Oliveri, Alberto; Cassottana, Luca; Laudani, Antonino; Fulginei, Francesco Riganti; Lozito, Gabriele Maria; Salvini, Alessandro; Storace, Marco

被引用次数: 17

2017

BAL: A library for the brute-force analysis of dynamical systems

COMPUTER PHYSICS COMMUNICATIONS

Linaro, Daniele; Storace, Marco

被引用次数: 2

2016

Open-Loop Compensation of Hysteresis and Creep Through a Power-Law Circuit Model

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Oliveri, Alberto; Stellino, Flavio; Caluori, Guido; Parodi, Mauro; Storace, Marco

被引用次数: 25

2016

Hysteresis and creep: Comparison between a power-law model and Kuhnen's model

PHYSICA B-CONDENSED MATTER

Oliveri, Alberto; Stellino, Flavio; Parodi, Mauro; Storace, Marco

被引用次数: 5

2016

Low-complexity digital architecture for solving the point location problem in explicit Model Predictive Control

JOURNAL OF THE FRANKLIN INSTITUTE-ENGINEERING AND APPLIED MATHEMATICS

Oliveri, Alberto; Gianoglio, Christian; Ragusa, Edoardo; Storace, Marco

被引用次数: 5

2015

Automatic Domain Partitioning of Piecewise-Affine Simplicial Functions Implementing Model Predictive Controllers

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Oliveri, Alberto; Reimers, Martin; Storace, Marco

被引用次数: 9

2015

A Circuit Model of Hysteresis and Creep

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Biggio, Matteo; Oliveri, Alberto; Stellino, Flavio; Parodi, Mauro; Storace, Marco

被引用次数: 16

2015

Efficient transient noise analysis of non-periodic mixed analogue/digital circuits

IET CIRCUITS DEVICES & SYSTEMS

Biggio, Matteo; Bizzarri, Federico; Brambilla, Angelo; Storace, Marco

被引用次数: 4

2015

Low-complexity piecewise-affine virtual sensors: theory and design

INTERNATIONAL JOURNAL OF CONTROL

Rubagotti, Matteo; Poggi, Tomaso; Oliveri, Alberto; Pascucci, Carlo Alberto; Bemporad, Alberto; Storace, Marco

被引用次数: 3

2014

Transient dynamics of an adiabatic NEMS

ANNALEN DER PHYSIK

Biggio, Matteo; Cavaliere, Fabio; Storace, Marco; Sassetti, Maura

被引用次数: 7

2014

Memory characteristics of hysteresis and creep in multi-layer piezoelectric actuators: An experimental analysis

PHYSICA B-CONDENSED MATTER

Biggio, Matteo; Butcher, Mark; Giustiniani, Alessandro; Masi, Alessandro; Storace, Marco

被引用次数: 13

2014

Accurate and Efficient PSD Computation in Mixed-Signal Circuits: A Time-Domain Approach

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Biggio, Matteo; Bizzarri, Federico; Brambilla, Angelo; Storace, Marco

被引用次数: 9

2014

Nonlinear behavioural model of charge pump PLLs

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Brambilla, Angelo; Linaro, Daniele; Storace, Marco

被引用次数: 17

2013

High-Speed Piecewise Affine Virtual Sensors

IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS

Poggi, Tomaso; Rubagotti, Matteo; Bemporad, Alberto; Storace, Marco

被引用次数: 16

2012

FPGA Implementations of Piecewise Affine Functions Based on Multi-Resolution Hyperrectangular Partitions

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Comaschi, Francesco; Genuit, Bart A. G.; Oliveri, Alberto; Heemels, W. P. Maurice H.; Storace, Marco

被引用次数: 20

2012

Codimension-Two Homoclinic Bifurcations Underlying Spike Adding in the Hindmarsh-Rose Burster

SIAM JOURNAL ON APPLIED DYNAMICAL SYSTEMS

Linaro, Daniele; Champneys, Alan; Desroches, Mathieu; Storace, Marco

被引用次数: 42

2012

Accurate and Fast Simulation of Channel Noise in Conductance-Based Model Neurons by Diffusion Approximation

PLOS COMPUTATIONAL BIOLOGY

Linaro, Daniele; Storace, Marco; Giugliano, Michele

被引用次数: 69

2011

Inferring network dynamics and neuron properties from population recordings

FRONTIERS IN COMPUTATIONAL NEUROSCIENCE

Linaro, Daniele; Storace, Marco; Mattia, Maurizio

被引用次数: 6

2011

Digital architectures realizing piecewise-linear multivariate functions: Two FPGA implementations

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Storace, Marco; Poggi, Tomaso

被引用次数: 36

2011

Ultra-Fast Stabilizing Model Predictive Control via Canonical Piecewise Affine Approximations

IEEE TRANSACTIONS ON AUTOMATIC CONTROL

Bemporad, Alberto; Oliveri, Alberto; Poggi, Tomaso; Storace, Marco

被引用次数: 87

2011

CONTINUATION ANALYSIS OF A PHASE/QUADRATURE ELECTRONIC OSCILLATOR

JOURNAL OF CIRCUITS SYSTEMS AND COMPUTERS

Bizzarri, Federico; Linaro, Daniele; Storace, Marco; Brambilla, Angelo

被引用次数: 3

2010

Integrated circuit implementation of multi-dimensional piecewise-linear functions

DIGITAL SIGNAL PROCESSING

Di Federico, Martin; Poggi, Tomaso; Julian, Pedro; Storace, Marco

被引用次数: 10

2010

Piecewise linear approximations of multivariate functions: A multiresolution-based compression algorithm suitable for circuit implementation

APPLIED NUMERICAL MATHEMATICS

Parodi, Mauro; Gaggero, Mauro; Storace, Marco

被引用次数: 4

2010

Digital Circuit Realization of Piecewise-Affine Functions With Nonuniform Resolution: Theory and FPGA Implementation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Poggi, Tomaso; Comaschi, Francesco; Storace, Marco

被引用次数: 23

2010

HARMONIC ANALYSIS OF OSCILLATORS THROUGH STANDARD NUMERICAL CONTINUATION TOOLS

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Bizzarri, Federico; Linaro, Daniele; Oldeman, Bart; Storace, Marco

被引用次数: 4

2010

The Hindmarsh-Rose neuron model: Bifurcation analysis and piecewise-linear approximations

CHAOS

Storace, Marco; Linaro, Daniele; de Lange, Enno

被引用次数: 192

2008

BIFURCATION ANALYSIS OF AN IMPACT MODEL FOR FOREST FIRE PREDICTION

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Bizzarri, Federico; Storace, Marco; Colombo, Alessandro

被引用次数: 5

2008

On the complexity of periodic and nonperiodic behaviors of a hysteresis-based electronic oscillator

CHAOS

Stellardo, Daniele; Bizzarri, Federico; Storace, Marco; De Feo, Oscar

被引用次数: 3

2007

学术合作



Alberto Oliveri

合作次数: 17

热那亚大学

Matteo Lodi

合作次数: 6

热那亚大学

Flavio Stellino

合作次数: 5

热那亚大学

Angelo Brambilla

合作次数: 5

米兰理工大学

Andrea Formentini

合作次数: 3

热那亚大学

还有 18 个合作者